NAME: ______________________________________ CLASS: ________ INDEX: _________

CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
Higher 2

BIOLOGY
Paper 4 Practical

9744 / 04
14th Aug 2018

2 hours 30 minutes

Candidates answer on the Question Paper

Additional Materials:

READ THESE INSTRUCTIONS FIRST

Write your index number and name on all the work you hand in.
Give details of the practical shift and laboratory, where appropriate, in the boxes provided.

Write in dark blue or black pen.
You may use a HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	
2	
3	
TOTAL	

Answer **all** questions

QUESTION 1

Diarrhoea in babies can be a serious problem that can have several different causes. One such cause is carbohydrate intolerance, usually stemming from a genetic condition which results in a very low level of a biomolecule **K**, which helps break down starch and sucrose in the intestine.

P is an extract of intestinal juices from a baby patient which needs to be tested for its concentration of **K**. You are tasked to help determine the concentration of the biomolecule **K** in extract **P**.

You are provided with the following:

1. The sample of intestinal juice from patient labelled **P**.
2. 10 cm^3 solution labelled **K1** containing 5.0% **K**.
3. 20 cm^3 solution labelled **S** containing 0.5% starch.
4. Distilled water, **W**.
5. Iodine solution.

*Before starting the investigation, read through steps **1-8** (including **(a)(i)**) and prepare a table in **(a)(ii)**.*

Proceed as follows:

1 Make a series of dilutions of **K** by **serial dilution**, diluting the concentration by a factor of 5.

2 You will need to repeat the dilution 4 times so that you have a range of 5 different concentrations. You are required to prepare for at least 5cm^3 of each concentration for step **3** onwards.

(a) (i) Complete the Table 1.1 to show how you will prepare the 5 different concentrations of **K**.

Table 1.1

Solution	Concentration of **K** / %	Volume of solution **K** taken from the previous dilution / cm^3	Volume of distilled water, **W** / cm^3	Total volume / cm^3
K1	5.0			

[2]

3 Set up a water bath with water between 38 to 42°C using the 500ml beaker. Monitor this temperature using the thermometer provided and maintain this temperature until you have completed step **8**.

4 Put 2 cm^3 of **S** into 6 separate test-tubes, then incubate these solutions in the water bath for at least 3 minutes.

5 Add 2 cm^3 of **K1** to one test-tube from step **4** and mix well. The reaction between **K** and **S** will begin as soon as **K1** is added.

6 Leave the solution to react in the water-bath for 5 minutes. Ensure that the solution mixture is mixed well throughout this 5 minutes.

7 After 5 minutes, add 0.5 cm^3 of the iodine solution into the test tube.

8 Repeat step **4** to **6** for all remaining concentrations of **K** you have prepared in step **1**. Record your results in **(a)(ii)**.

(ii) Record your results in a suitable table in the space below.

[3]

9 Repeat step **4** to **6** for solution **P** and using your results from step **7**, estimate the concentration of **K** in the intestinal juice of the patient.

(iii) Estimation of concentration of **K** in sample **P**: ……………………………………………………….[1]

(iv) Hence, based on this concentration of **K** you have estimated, conclude with justifications whether the patient from which **P** is extracted is likely to have the genetic condition.

……

………………………………………………………………………………………………………[1]

(v) Given that the concentration of **K** for a patient with the genetic condition is around 0.05%, comment on the accuracy of your results in part **(a)(ii)** in helping you determine if the patient from which **P** is extracted has the genetic condition.

……

……

……

………………………………………………………………………………………………………[2]

(vi) Suggest how the procedure from steps **1** to **2** could be modified to increase the accuracy of your results.

………

………[1]

(vii) From the results of your investigation, deduce with rationale the identity of **K**, and suggest a control that you can perform to help you determine with more certainty the identity of **K**.

………

………

………

………[2]

(viii) Apart from ensuring that the solution mixtures were mixed well, identify two other aspects of the design of this investigation and explain how they are important in allowing valid comparisons to be made between the results in **(a)(ii)**.

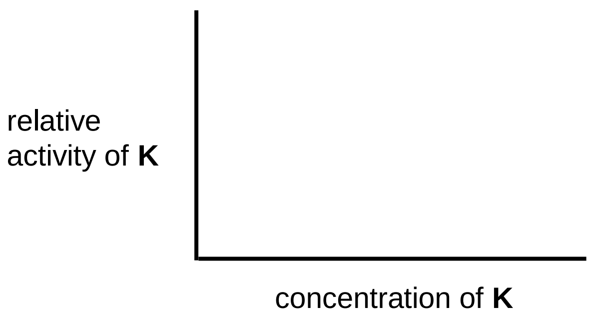
………

………

………

………[2]

(ix) The results of your investigation in **(a)(ii)** have allowed you to make qualitative comparisons between the various tubes. In order to complete a graph such as that shown below, it is necessary to obtain numerical values for the y (vertical) axis which give an indication of relative activity of **K**.



Outline how the procedure for this investigation could be further modified in order to obtain quantifiable numerical values for the relative activity of **K**.

………

………

………

………[2]

[Turn over

(b) A student carried out such a procedure to obtain quantifiable numerical values for the relative activity of **K**, using a similar procedure in **(a)** and measuring the intensity of the positive results of the starch-iodine solution test. The results for this investigation is shown in Table 1.2.

Table 1.2

Concentration of **K** / %	Relative intensity of starch-iodine solution after 5 minutes / AU (arbitrary unit)
5.0	0.20
4.0	1.55
3.0	2.05
2.0	2.90
1.0	3.55

(i) Use the grid below to display the student's results obtained in **(b)(i)** in an appropriate form.

[4]

(ii) Another patient B's intestinal juice sample produced a relative intensity of 3.20 AU with the starch-iodine solution test.

Show clearly how you would use your answer in **(b)(i)** to estimate the concentration of **K** in this patient.

……….

……….[1]

(iii) After setting up a negative control, the student measured a result of 4.00 AU (arbitrary unit) with the starch-iodine solution test. Assuming that the negative control resulted in 0.0 unit of relative activity of **K**, calculate the estimated relative activity of **K** in patient B. Show your working clearly.

[2]

[Total: 23]

QUESTION 2

During this question you will require access to a microscope and slide **S1** and **S2**.

In this question, you will be investigating how cells appear during cellular division for different organisms.

(a) Slide **S1** shows a stained longitudinal section of a young root tip.

Examine **S1** carefully using low and high power objectives of your microscope. Note the occurrence and distribution of different kinds of cells in this section.

(i) Make a plan drawing of the entire section, to show the different **regions** of the root tip. These regions result from differences in the *shapes, sizes* and *structure* of the cells as well as in the frequency of mitosis. Do not draw individual cells.

Label in your diagram where the cap of the root tip is, and annotate your drawing to show the relative mitotic activity in each region that you map.

[4]

(b) Slide **S2** shows a unicellular eukaryote, *Paramecium*, that are usually found in freshwater bodies. They have two nuclei, a micronucleus that stores most of the genome, and a macronucleus which expresses the genes for daily functioning. Paramecium reproduces asexually, by binary fission. During reproduction, the macronucleus splits by a type of amitosis, and the micronuclei undergo mitosis. The cell then divides transversally, and each new cell obtains a copy of the micronucleus and the macronucleus.

(i) Look at slide **S2** under the high power objective. In the space below, make a drawing of a *Paramecium* that is clearly undergoing binary fission, where you can see two cells dividing from one. In your diagram, label one macronucleus as well as the line of divide.

[5]

(ii) Locate a cell undergoing **anaphase** in **S1**. Using a suitable table, record any observable differences you have noticed between the cell you have located in **S1** and the cell you have chosen to draw in **(b)(i)**.

[2]

(iii) Using the stage micrometer to calibrate your microscope, show with working how much each eye piece graticule measures at objective lens of 40X.

[2]

(iv) Using your answer in **(iii)**, measure the size of the cell that you have drawn in **(i)** and show the magnification of your drawing.

[2]

(c) In a recent nitrate pollution of the school pond near the science laboratory, there was an increase growth in algae. A student decided to investigate whether the population size of *Paramecium* was affected by this pollution. In his investigation in growing *Paramecium* in two different types of water, he obtained a set of results shown in **Table 2.1**.

Table 2.1

Mean population size per 100 ml / cell count		Significance of difference	Total number of samples
Normal pond water	Polluted pond water		
530	891	$p < 0.05$	10

(i) State a statistical test that could be used to determine if the difference between the mean population sizes of *Paramecium* in the two different water bodies is significant.

………[1]

(ii) Based on the result seen in Table 2.1, explain what conclusions can be drawn for the results obtained by the student for the effect of nitrate pollution.

………

………

………

………[2]

[Total: 18]

QUESTION 3

Bio-indicators refer to organisms being used to measure changes in the environment such as pollution level or ambient temperature, mainly because of how biotic factors can be taken into consideration which traditional instruments are not able to detect. Insects are a group of organisms that are frequently used as bio-indicators as they are abundant and are highly sensitive to even small changes in the environment due to their small sizes and natural morphologies. As insects have a very narrow physiological tolerance, climate change is going to affect these organisms even more so than other larger organisms.

As ectotherms (cold-blooded organisms), the rate of metabolism in insects are highly dependent on ambient temperature. The rate of respiration can be used to estimate the level of metabolism in insects. You are required to design an experiment to investigate how temperature affects the relative metabolic rates of crickets, an insect commonly found in Singapore.

Figure 3.1 shows an example of a set-up for a simple respirometer measuring respiration for seeds. Show in your plan how you would modify this set-up to measure the rate of respiration in crickets.

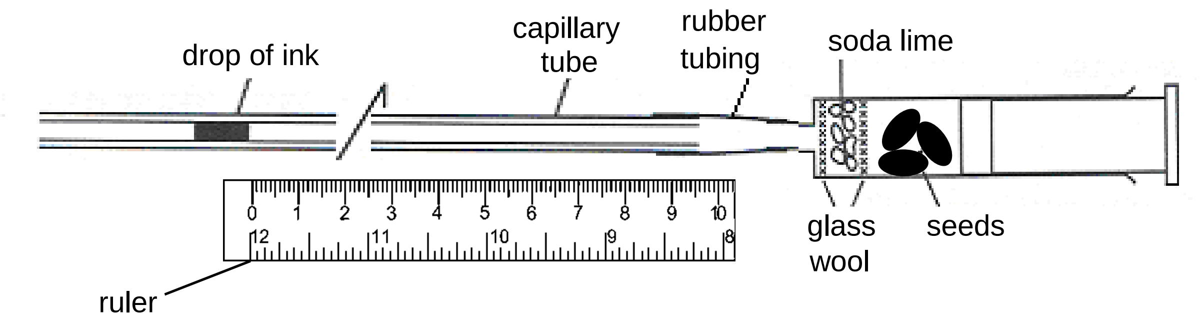


Fig. 3.1

In your plan, you **must** use:
- crickets
- soda lime
- delivery tube
- thermometer
- ruler

You may select from the following apparatus and use appropriate additional apparatus:
- normal laboratory glassware, e.g. test-tubes, boiling tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.
- syringes
- Bunsen burner
- timer, e.g. stopwatch

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
- be illustrated by relevant diagrams, if necessary, to show, for example, the arrangement of the apparatus used
- identify the independent and dependent variables
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible
- include layout of results tables and graphs with clear headings and labels
- use the correct technical and scientific terms
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 14]

End of Paper